

RONIT NAIK

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Education

Northeastern University

Sep 2024 - Apr 2026

Master's in Computer Science — Artificial Intelligence — GPA: 3.84

Boston, Massachusetts

- Courses: Algorithms, Programming Design Patterns, Cloud Computing, LLMs with Graph DB, Computer Vision

Vellore Institute of Technology

Sept 2020 - May 2024

Bachelor's of Technology in Computer Science with AI and ML — GPA: 8.88

India

- Courses: Data Structures & Algorithms, Web Programming, Design Patterns in Java, Operating Systems, Python

Work Experience

AstraZeneca

Jun 2025 – Dec 2025

Data Science and Data Management Co-op

Wilmington, DE

- **Reduced duplicate support queries by 40%** by engineering an automated FAQ system using Gemini-2.5-Flash and **Sentence-Transformers**, utilizing **DBSCAN** clustering and semantic similarity matching.
- **Uncovered 12 critical data inconsistencies** in Customer Engagement Data Products by building a production-ready **RAG pipeline** with **Airflow**, **Databricks Genie** and **AWS Athena** to translate natural language queries to SQL.
- **Enhanced downstream analytics accuracy** by developing an **LLM-powered address-matching pipeline** to standardize, clean, and deduplicate complex Healthcare Professional (HCP) and Healthcare Account (HCA) records.
- **Led exploratory data analysis (EDA)** on large-scale customer engagement datasets using **AWS Athena** and **Python**, identifying key behavior patterns to perform executive **stakeholder reporting** and pipeline architecture.

Northeastern University

Sep 2024 – Jun 2025

Graduate Teaching Assistant

Boston, Massachusetts

- Conducted weekly office hours, guided **400+** students to build better projects, and graded assignments
- Delivered detailed feedback, significantly boosting the understanding of **advanced OOPS and AI concepts**.

Quest Global

Jan 2024 – May 2024

Machine Learning - Software Engineer Intern

India

- **Achieved 95% precision in production** for anomaly detection models by deploying them on **AWS SageMaker** and architecting a serverless inference pipeline using **AWS API Gateway** and **AWS Lambda**.
- **Streamlined model development and deployment cycles** by engineering robust **ETL pipelines** and leveraging **MLflow** for end-to-end model versioning and experiment tracking.
- **Accelerated data visualization and model integration** by enhancing a full-stack enterprise web application utilizing **Spring Boot**, **ReactJS**, and **PostgreSQL**.

Projects

IncidentStream: Real-Time Incident Alert Dashboard

Terraform | AWS | Node.js | React.js

- Engineered a real-time incident monitoring platform with a serverless **SNS** → **SQS** → **Lambda** pipeline, pushing severity-scored alerts to clients via **WebSocket** in under 1 second.
- Provisioned **36 AWS resources** across **6 Terraform** modules (**VPC**, **DynamoDB**, **SNS/SQS**, **Lambda**, **API Gateway**, **WebSocket**, **S3/CloudFront**), implementing modular **IaC** with least-privilege **IAM** policies.
- Developed a **React** dashboard with live **WebSocket** integration featuring real-time alert streaming, one-click acknowledge/escalate/resolve actions, severity filtering, and a stats panel tracking **MTTA** metrics.

Intelligent Job Application Tracker with LinkedIn Auto-Scraping

FastAPI | Kafka | PostgreSQL

- Built an event-driven job tracker with **FastAPI**, **Kafka**, **Redis**, and **PostgreSQL** processing 100+ applications with **Chrome extension** for **LinkedIn scraping** and AI-powered resume-job matching analyzing **50+** skill categories.
- Implemented caching strategy reducing API latency by **60%** and costs by **70%**, featuring **JWT Auth**, **Docker**, and **15+ RESTful** endpoints.

Cryptocurrency Price Prediction with Twitter Sentiment Analysis

TensorFlow | Keras | Python

- Engineered a multi-modal pipeline using **Apache Airflow** to orchestrate live Twitter sentiment and historical data.
- **Developed a TensorFlow NLP** pipeline to extract text sentiment with **ARIMA** models for automated trading signals.
- **Streamlined the deployment cycle** by containerizing the system with **Docker** and leveraging **MLflow** for robust experiment tracking, model versioning, and hyperparameter tuning.

Technical Skills

Languages: Python, Java, SQL, HTML5, CSS, JavaScript, C, C++, Cypher, ReactJS, React-Native

Developer Tools: AWS, Terraform, Postman, PowerBI, Git, Neo4j, Azure, Docker, Kafka, Jira, MLflow, Airflow

Libraries/Frameworks: Tensorflow, PyTorch, Spring Boot, PostgreSQL, Express.js, LangGraph, LangChain, Node.js